

01 · UK-DALE — four years of one London home

Dataset: UK-DALE (Kelly & Knowlton, *BuildSys* 2015, CC-BY 4.0) — five London homes, 2012–2017, held here as raw-fidelity Parquet under ``data/fnd/ukdale/``. **Role:** the primary training/evaluation substrate for appliance-level disaggregation.

> How to read: every section below starts from a question, answers it from the raw files, then interprets
> the answer. Charts are rendered — read top-to-bottom without executing anything. No number is hard-coded.

```
python: 3.13.7 | pandas 3.0.6 | numpy 2.5.3
repo: /root/longvdh/agent_space/projects/watt-wiser
```

Meet the dataset

UK-DALE recorded **five real London homes between 2012 and 2017** for exactly the problem this project studies: appliance-level electricity from meter data. Two instrument classes were installed:

- a meter on the **incoming supply** recording total active power **every second** (houses 1, 2 and 5; house 1 also logs apparent power and voltage),
- small meters on **individual appliances** — a fridge here, the kettle there — recording roughly **every 6 seconds**.

What we hold is the raw recording, unmodified: one Parquet file per meter (``ts_us`` = UTC microseconds, ``v0`` = active watts), a ``labels.dat`` mapping channel numbers to appliance names, and sparse "button press" interaction logs. Nothing resampled, nothing cleaned. The table below is computed from Parquet metadata only.

Why this dataset matters to this project: in the target deployment only the aggregate is collectable. Recordings like these - with appliance-level ground truth - are how we learn what the aggregate is made of and what a model is being asked to infer. Sections tagged **[insight only]** exist for that understanding; **[collectable]** marks what we can record ourselves.

house	channels	mains	span	channel rows
house_1	53	yes	4.5 y (from 2012-11)	728,877,165
house_2	19	yes	0.6 y (from 2013-02)	40,506,387
house_3	5	no	0.1 y (from 2013-02)	2,581,158
house_4	6	no	0.6 y (from 2013-03)	13,088,208
house_5	25	yes	0.4 y (from 2014-06)	33,903,991

House 1 is the flagship — 4.1 years of mains plus 53 labelled channels — so this notebook walks through it.

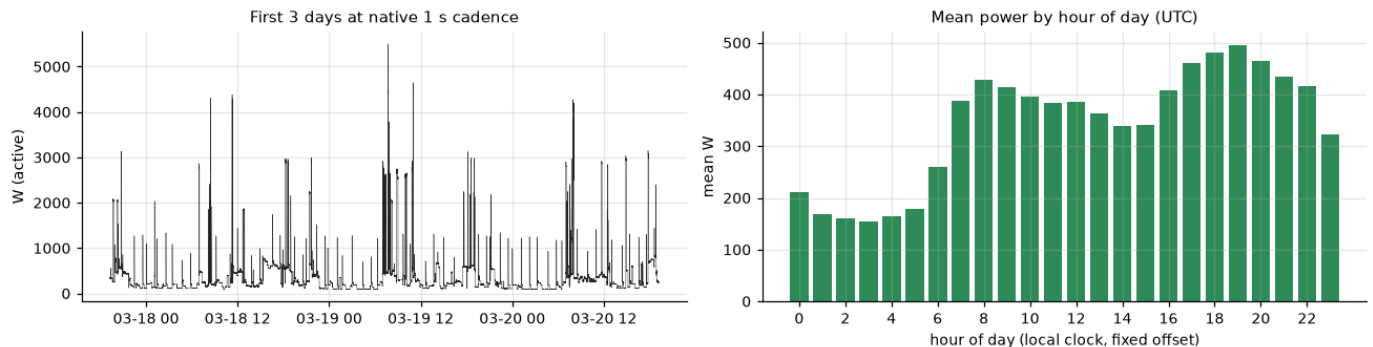
A day in the life

What does the aggregate signal of a real home actually look like?

The supply meter wrote **128 million one-second readings** between 2013-03 and 2017-04 — 12,268 kWh of electricity. The scan below streams every row once (quantiles from a uniform 2M-sample decimation); then we look at three days up close and at the average day.

```
series      : UK-DALE house_1 mains (active W @ 1 s)
rows        : 128,238,700 (missing 0)
span        : 1500.9 days
cadence     : median 1.00 s, p90 1.10 s, gap fraction 0.000
power W     : mean 342.5 | p50 223.3 | p95 693.3 | p99 2710.7 | max 8066
energy      : 12268.0 kWh
steps per day : >=30 W 1542.3 | >=100 W 709.9 | >=300 W 179.3
```

weekday mean 346 W | weekend mean 333 W | weekend/weekday 0.96 (whole span, UTC days)



Reading it. The load never switches off: an overnight floor around 150–160 W (standby electronics plus the fridge ticking over) rises through the morning, and the evening peak reaches ~500 W on average around 19–20 h. On top of that floor sit short spikes of 2–5 kW — kettle boils, a shower, the hob — lasting seconds to minutes. The mean (342 W) sits far above the median (223 W) and p99 is 2.7 kW: demand is bursts, not a steady draw. This asymmetry is why a mean alone never summarises a home. Two more structural reads: weekends run a few percent lighter than weekdays (printed above), and every hour-of-day statistic here is **UTC** — add one hour for British Summer Time, which applies to the zoom windows below (late April).

Q1 — Is anything missing?

Before trusting any statistic: what did the logger actually miss?

One streaming pass over the mains finds every interval longer than a minute between consecutive readings — and, as a by-product, the energy of every calendar month (reused in Q5).

```
gaps > 60 s: 21 | total missing 18.5 d of 1500.9 d (1.23%)
gap length  starts (UTC)
-----
245.82 h    2016-07-21 19:51
102.12 h    2016-03-24 21:08
43.66 h     2013-04-01 21:12
21.52 h     2013-04-11 12:11
9.82 h      2014-07-21 17:29
7.74 h      2015-05-01 19:13
5.92 h      2014-03-03 19:52
2.66 h      2013-04-15 11:39
```

Reading it. The recording is excellent for its age — but it is not gapless: **21 gaps over a minute in 4.1 years, ~18.5 days of missing time (1.23%)**. Two long holes dominate (~10 days from 2016-07-21, ~4 days from 2016-03-24), plus a 2-day hole in April 2013 and a scatter of multi-hour evening dropouts. Any monthly statistic for 2016 must be read with the holes in mind; Q5 flags the affected months on its chart.

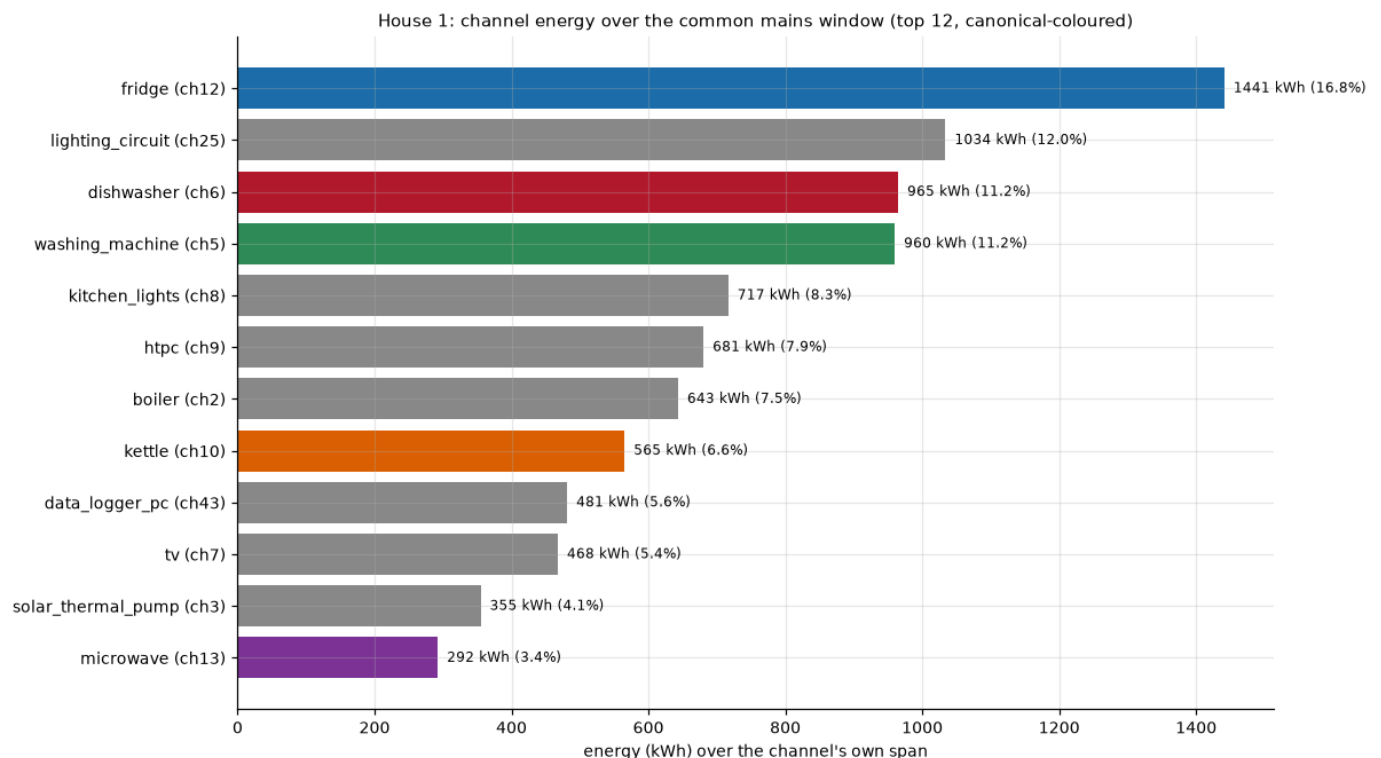
Q2 — Which appliances matter?

Of the 53 labelled channels, which ones carry the energy?

For every channel we compute the project's ON rule (``thr_on = max(5 W, 0.5 × median of samples above the 5 W noise floor)`` — the same formula as the gold layer), duty cycle, episode counts and dwell percentiles. Energy is integrated with real timestamps **over the same window as the site meter**, with each sample's carry-over across a gap capped at 3× the channel's cadence, so channel and site numbers are directly comparable; channels are also flagged by the fraction of the window they actually record. House 1's channels idle near 0 W, so the plain ON rule is safe here — a floor-aware variant appears in the cross-house table below. The full read of $53 \times \sim 20\text{M}$ rows takes a minute or two; the table shows the top consumers.

ch	label	dt_med_s	p50_on_W	thr_on_W	duty_%	kWh	rec_%	eps/day	dwell_p50_s	dwell_p90_s
1	aggregate	6.0	256.0	128.0	100.0	13834.0	99.1	0.2	30.0	544.8
12	fridge	7.0	89.0	44.5	42.4	1441.2	99.0	25.3	1456.0	2464.0
25	lighting_circuit	6.0	64.0	32.0	24.3	1034.0	99.1	77.6	18.0	186.0
6	dishwasher	7.0	121.0	60.5	2.6	964.8	99.2	1.8	931.0	2611.0
5	washing_machine	7.0	177.0	88.5	3.4	959.9	98.3	19.4	21.0	385.0
8	kitchen_lights	6.0	130.0	65.0	13.4	716.7	98.4	12.6	192.0	2708.4
9	htpc	7.0	69.0	34.5	24.4	680.5	99.3	2.5	4242.0	17444.0
2	boiler	6.0	12.0	6.0	87.9	643.3	97.9	42.0	24.0	720.0
10	kettle	7.0	2347.0	1173.5	0.7	565.3	96.9	4.8	112.0	189.0
43	data_logger_pc	7.0	13.0	6.5	100.0	481.3	99.3	0.0	65045365.0	83500282.6
7	tv	7.0	103.0	51.5	11.6	467.7	99.3	1.6	5383.0	13267.8
3	solar_thermal_pump	6.0	48.0	24.0	20.8	354.8	97.9	8.5	168.0	5527.2
13	microwave	7.0	1523.0	761.5	0.5	292.4	99.0	6.5	35.0	133.0
11	toaster	7.0	1576.0	788.0	0.5	278.6	98.7	2.3	196.0	231.0
18	adsl_router	7.0	6.0	5.0	95.4	251.1	97.4	9.0	1099.0	10150.0

energy: timestamp-integrated over the mains window, per-sample carry-over capped at 3x cadence
 channels recording < 5% of the window (energy covers recorded time only): ch22 Hoover 0.7%, ch39 hair_dryer 1.0%, ch40 straighteners 0.2%, ch41 iron 0.0%, ch20 soldering_iron 2.4%, ch30 DAB_radio_livingroom 3.2%

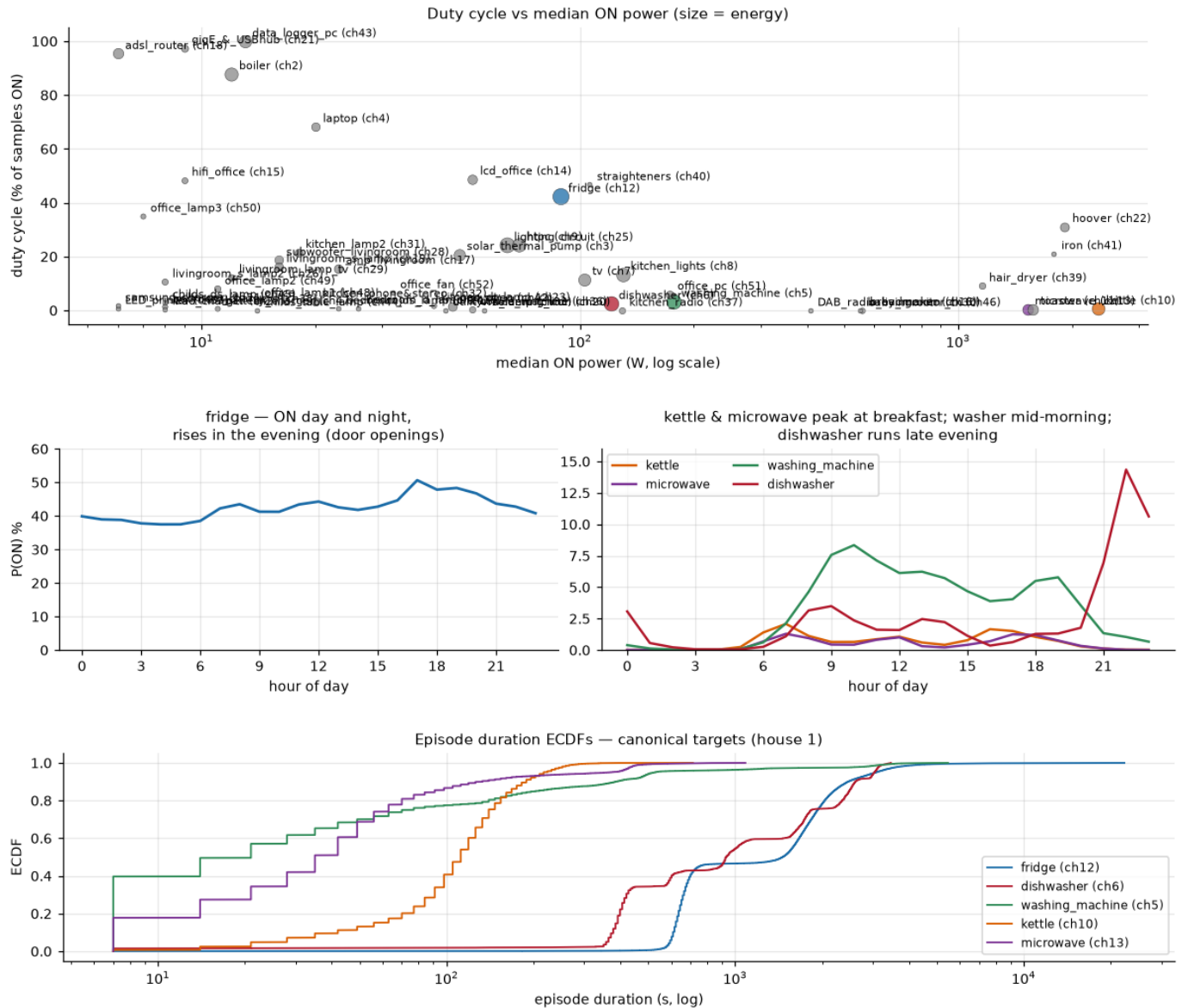


Reading it. The fridge is the single biggest labelled consumer ($\approx 1,440$ kWh over the mains window), with the dishwasher (≈ 965) and washing machine (≈ 960) in a near tie behind it. The next tier is **not** appliance-shaped: lighting (two channels together $\approx 1,750$ kWh), an always-on home-theatre PC (≈ 680), the boiler circuit (≈ 640). The five canonical targets account for **34%** of the whole home's energy (their $\approx 4,220$ kWh over the site meter's 12,268); lights and electronics form the next tranche; the rest is a long tail. Note the table's top row: channel 1, the 6-second "aggregate", integrates $\approx 13,800$ kWh — 13% more than the site meter over the same window (reconciled in Q7). Energy tells us what pays; episode counts

tell us what is *recognisable* — both views matter.

Q3 — How does each appliance behave?

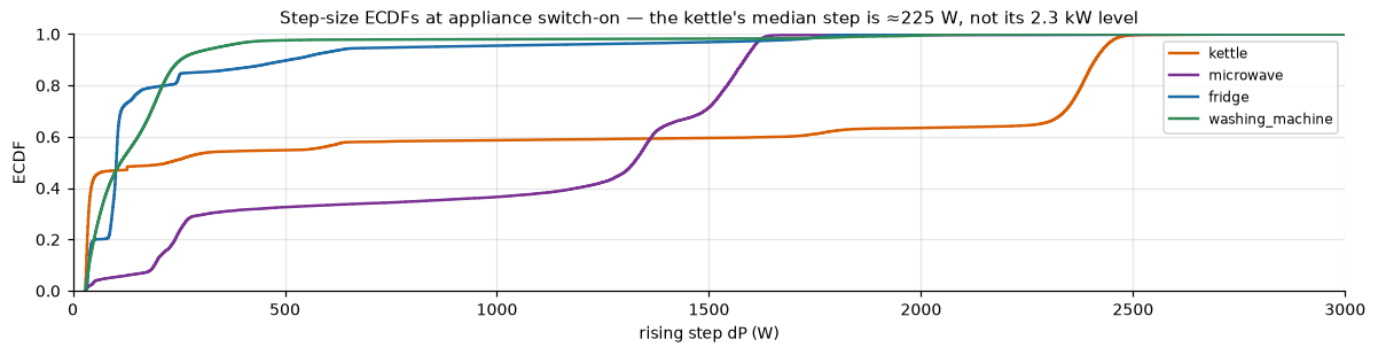
What does "ON" look like for each target — how often, how long, and when in the day?



How big is the step when a device switches on?

Episode counts say how often; the *step* the aggregate sees at each onset is what a disaggregator must detect. For every rising transition of at least 30 W (transitions spanning a sampling gap are dropped) of the canonicals plus two contrast appliances:

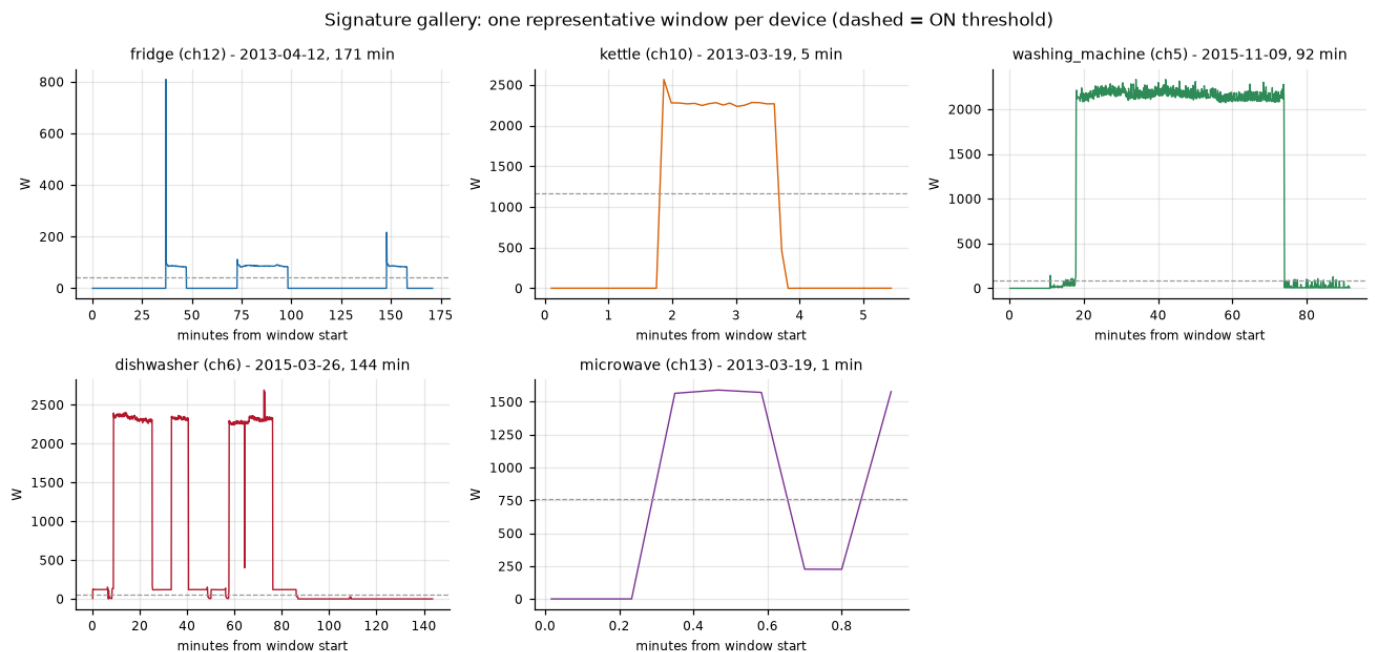
appliance	ch	median ON W	n steps	dP p50 W	dP p90 W
iron	41	1790	890	1778	1823
toaster	11	1576	6029	1545	1617
microwave	13	1523	20946	1323	1584
kettle	10	2347	17677	225	2411
washing_machine	5	177	252502	113	254
fridge	12	89	53257	102	511
dishwasher	6	121	25247	38	418



Reading it. The kettle draws 2.3 kW at its median ON level (Q2), yet its **median rising step is only ≈ 225 W** (p90 ≈ 2.4 kW): the thermostat cycles the element, so a boil arrives as a burst of moderate steps with occasional full-size ones — roughly 2.7 qualifying steps per boil. The iron is the opposite: a clean ≈ 1.8 kW step every time (p50 1,778 / p90 1,823). The fridge steps in ≈ 100 W compressor increments. For the target deployment this is the difference between threshold logic that works (iron) and one that silently misses (kettle) — step size is a property of the appliance, not of the meter.

What does the raw trace of each device look like?

Summaries are compact, but the raw trace is what a disaggregation model actually has to recognise. One representative window per device - for the burst appliances (kettle, microwave, fridge) the episode closest to the median duration; for the washer the longest episode (its median "episode" is a 14-second control draw); padded for context, with each device's ON threshold drawn dashed.



Reading it. The gallery makes the signatures concrete: the fridge's compressor sawtooth with a periodic defrost spike; the kettle's single clean trapezoid at 2.3 kW; a real washing-machine program - 2.2 kW of heater and motor for well over an hour, not the 14-second control draws that dominate its episode count; the dishwasher's three heater pulses spread across two hours; the microwave's one-minute trapezoid. Even at 6-second cadence these shapes are distinct - that is what a disaggregation model keys on. The two circuits that never really switch off - the lighting circuit and the boiler - get their own exhibit below.

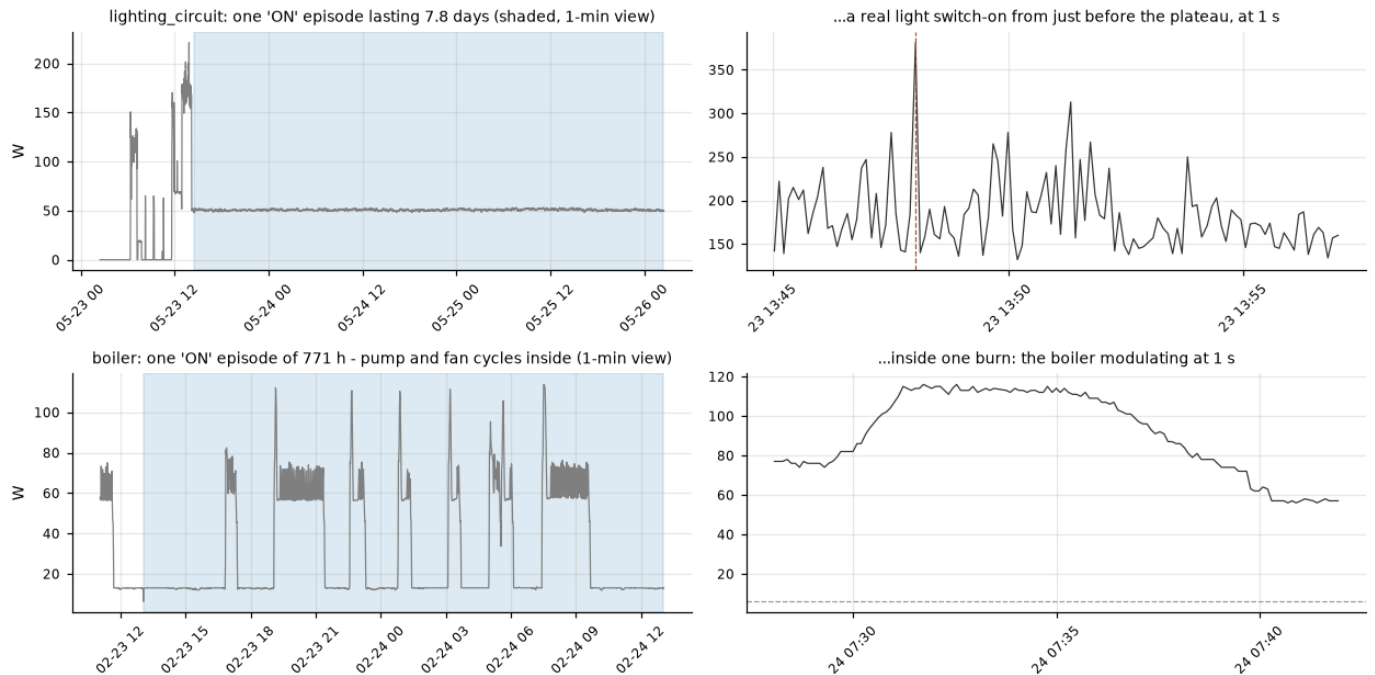
The scatter then separates appliance **types**: fridge = low power (89 W median ON) / high duty (42%); kettle = 2.3 kW / ON 0.7% of the time; washer and dishwasher = mid power, rare but long. The hour-of-day curves are the household's schedule made visible (hours UTC; British Summer Time would shift them one hour later on the local clock): **kettle and microwave at 7-8 h (breakfast), washing machine mid-morning (9-11 h), dishwasher late evening (21-24 h, peaking at 14%),** and the fridge on around the clock with an evening rise. Dwell ECDFs: kettle ≈ 2 min; dishwasher ≈ 16 min

median, $p90 \approx 1$ h. The washer's meter also catches **14-second control draws** (19.4 "episodes"/day) between real cycles — episode filtering needs a dwell floor.

The long runners - lighting_circuit and the boiler - [insight only]

Not every circuit blinks on and off. Two house 1 circuits stay effectively ON for days: the lighting circuit and the boiler (its pump and controller cycling almost continuously). The gallery showed bounded signatures; these need the opposite treatment - zoom out to see the plateau, then zoom in to see the switch and the pump.

lighting longest ON: 7.8 days | boiler longest ON: 771 h



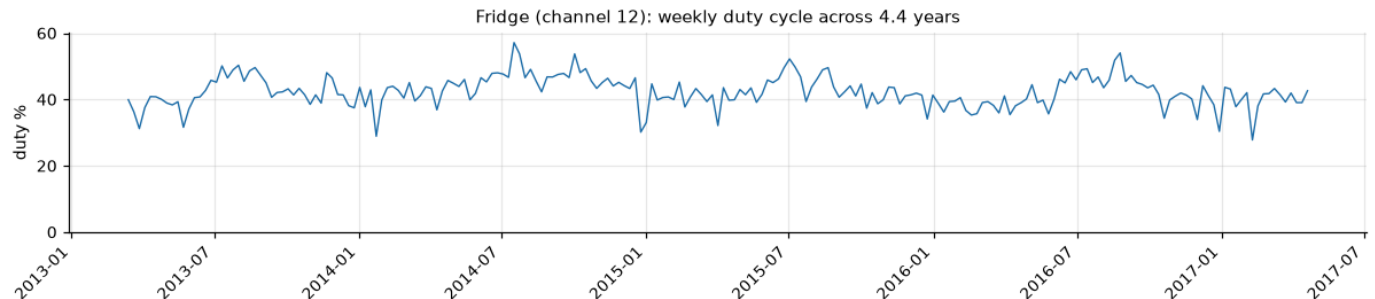
Reading it [insight only]. 'Appliance off = 0 W' is a fiction. The lighting circuit's longest 'ON' episode is a flat ~50 W plateau held for 7.8 days - an always-on device riding on the lighting circuit, not lights; real light switching (right panel) is a bursty step on top of such baselines. The boiler circuit idles at ~14 W and burns in smooth 60-115 W modulations, hundreds of times over a multi-week stretch - invisible individually in a daily mean, collectively most of its 664 kWh. Baselines are made of circuits like these; a model that only hunts big square pulses will miss them.

Q4 — Is the fridge stable across 4 years?

The fridge is the always-there anchor a model sees every day — does its behaviour drift?

Weekly duty cycle over the whole span:

fridge weekly duty: mean 42.4% | p5 35.0% | p95 49.6% | weeks 215
defrost-band (200-300 W) share of fridge ON time: 2.9% - a third state between compressor and OFF



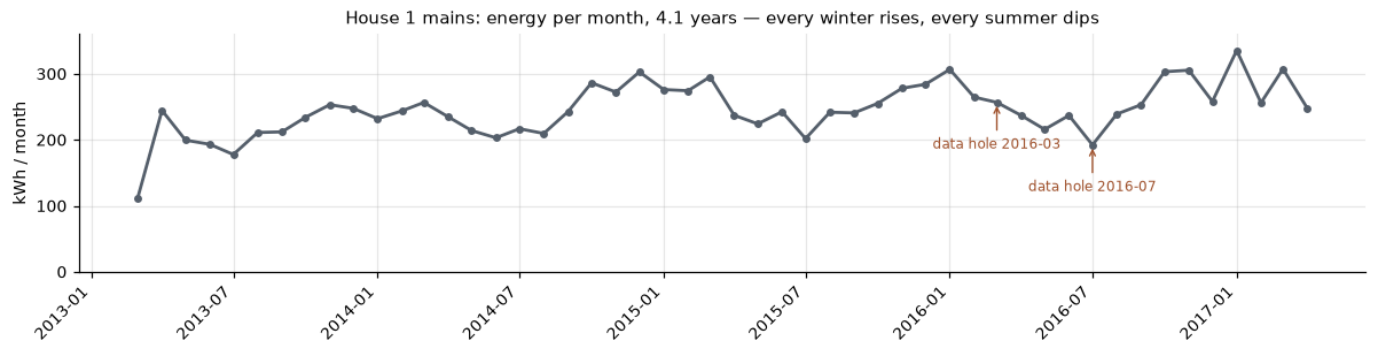
Answer: yes — stationary. Duty wobbles inside a 35-50% band (mean 42.4%) with no upward or downward trend across 215 weeks: the signature learned in year 1 still holds in year 4. Brief dips to ~30% recover within a week or two. One wrinkle the duty number hides: the compressor is not two-state — **2.9% of ON time sits in a 200-300 W defrost band** between the ~89 W compressor level and OFF (printed above), a third state a two-state model will systematically misread. This is the best-case appliance for signature learning.

Q5 — Is household consumption seasonal?

4.1 years is long enough to ask whether the house itself changes with the seasons.

Energy per calendar month (from the same streaming pass as Q1), with the two data-hole months marked:

winter (Nov/Dec/Jan) mean 279 kWh | summer (Jun/Jul) mean 208 kWh | ratio 1.34

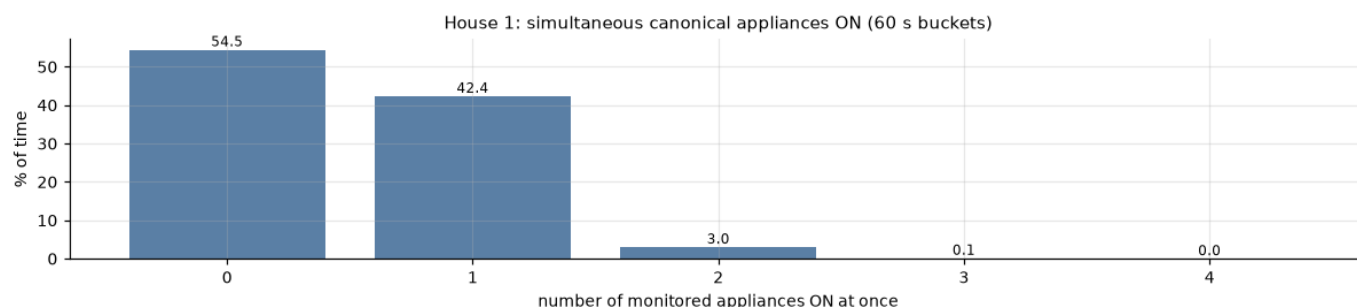


Answer: mildly but repeatably seasonal. Every winter rises to ~250-335 kWh/month, every summer dips to ~180-240 — winter is $\approx 1.35\times$ summer (279 vs 208 kWh on average), and the pattern repeats all five years. The two visible outlier dips (Mar 2016, Jul 2016) are the **data holes from Q1**, not behaviour — which is why Q1 comes first. Consequence: train/test splits that span seasons will see different baselines.

Q6 — Do appliances run together?

How much do the five canonical targets overlap? If they never overlapped, disaggregation would be trivial arithmetic.

share of 60 s buckets with 2+ ON: 3.2%



Reading it. 3.2% of 60-second buckets have two or more canonical appliances ON. A small number with a big consequence: the fridge alone is ON 42% of the time, so almost every kettle boil or wash cycle happens *on top of* a compressor cycle. The mains is a **superposition**, not a sum of disjoint events — disaggregation has to un-mix, not partition.

Event view: when a device starts, what else is already running?

The bucket view counts *time*. Disaggregation errors happen at *events* — so for every canonical episode of at least 30 s, we check which other canonicals were ON at its midpoint (nearest sample within 45 s):

appliance	episodes >= 30 s	% starting with another canonical ON	most frequent partner
-----	-----	-----	-----
fridge	37951	6.7	washing_machine (3%)
dishwasher	2637	48.0	fridge (45%)
washing_machine	23026	51.0	fridge (48%)
kettle	6619	52.0	fridge (46%)
microwave	3774	56.1	fridge (50%)

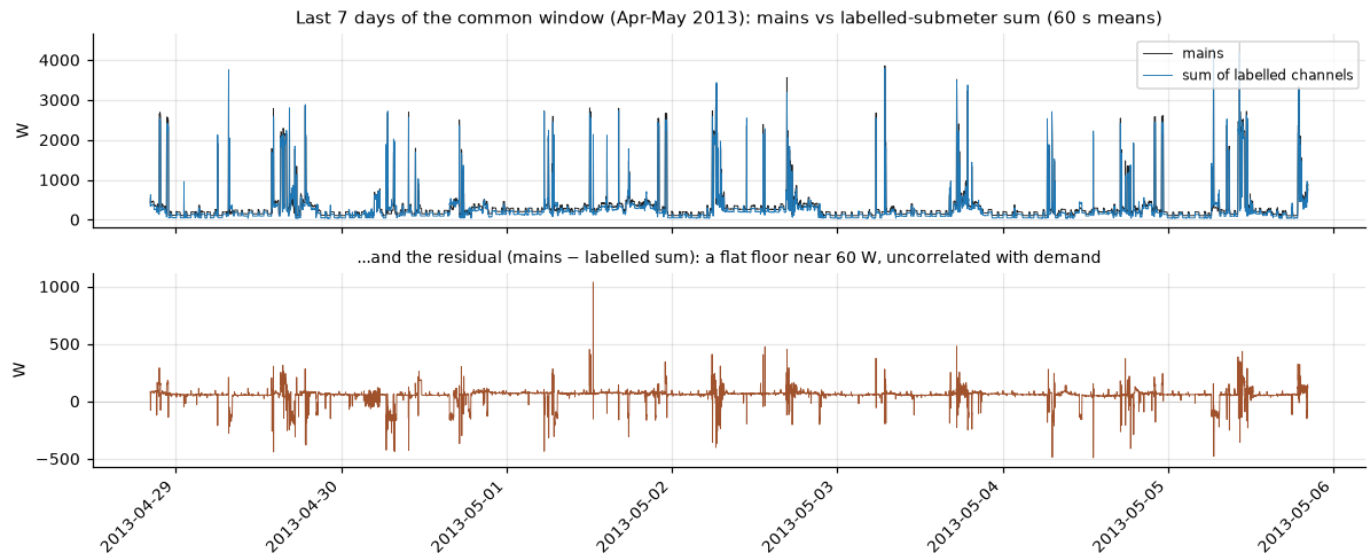
Reading it. Half of all kettle boils (52%), wash cycles (51%) and dishwasher runs (48%) begin while another canonical appliance is already running — most often the fridge, which is ON 42% of the time. The microwave is worst (56%). Compare the two views: only 3.2% of *minutes* have 2+ appliances ON, but roughly half of *events* start inside a mix. A disaggregator evaluated on events lives in the superposition; the bucket view alone flatters it.

Q7 — How much of the bill do we actually understand?

If we add up all labelled channels, how close do we get to the site meter?

Method: mean-pool the mains and every channel onto a common 60 s grid over a **30-day window in which all 52 labelled channels are active** (it ends 2013-05-05 — several channels stop recording soon after), then compare energies. Coverage sums only minutes where the site meter itself recorded, and the same window yields the **residual** (mains – labelled sum) shown below. Channel 1 is the aggregate itself — a label collision in `labels.dat` — and is excluded from the sum.

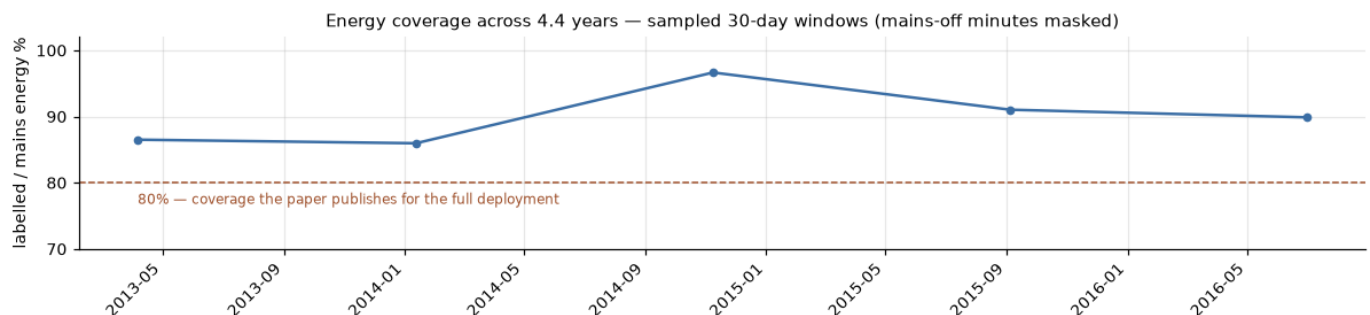
```
energy coverage (sum of 52 labelled channels, aggregate excluded / mains, last 30 days, mains-off minutes masked): 86.5%
r(labelled sum, mains) = 0.975 | residual mean 44.9 W (p25 56, p75 70) | night(02-05 UTC) 58.0 W, day 53.3 W
residual < 0 in 8.9% of minutes | r(residual, mains) = -0.060 | nightly mains floor (02-05 UTC): 111 W
channel_1 ('aggregate') vs site mains over the same window: 271.4 vs 238.7 kWh (13.7% apart)
```



Coverage over four years - and what the residual is made of

One window is a sample; the question is whether coverage holds across 4.4 years and what the leftover actually is. Four more 30-day windows (mains-off minutes masked — in the July 2016 hole the submeters kept logging while the site meter was down, and an unmasked ratio would exceed 100%):

```
2013-04-05: 86.5% | 2014-01-11: 86.0% | 2014-11-08: 96.7% | 2015-09-04: 91.0% | 2016-06-30: 89.9%
```



Answer: ≈87% here, 86-97% everywhere we sample — and the leftover is the meters, not the house. The labelled channels track the site meter closely ($r = 0.975$), but the remaining ≈13% is not hidden household load: the residual (bottom panel) is a **flat floor of ≈55-70 W** — night 58 W, day 53 W, $r(\text{residual, mains}) = -0.06$ — a constant offset, not a load that follows activity. That matches the ~50 W self-draw of the 52 battery-powered plug meters documented in the UK-DALE paper: the IAMs eat their own measurement. Two lessons. **[insight only]:** house 1's UNKNOWN class is mostly instrumentation, so its size must not transfer to our deployment — a Shelly install's residual will be genuinely unmonitored appliances. **[collectable]:** the method transfers — every deployment needs the same mains-vs-sum books-closing with an explicit **UNKNOWN** class; what is genuinely always-on here (the ≈111 W nightly mains floor, printed above) is a separate question. Note also that sampled windows

stay between 86% and 97% with no decay, while the paper publishes 80% for the full deployment — window choice moves the number more than any appliance decision, and channel 1's "aggregate" reads $\approx 13.7\%$ high over the same window (printed above), so the site meter is the canonical reference.

Q8 - What else is in the files? (and can we collect it?)

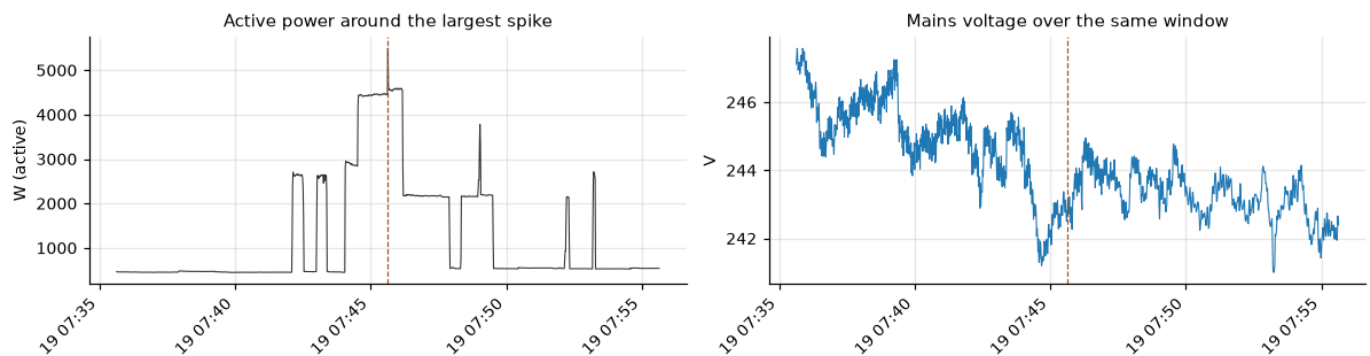
Everything above used active power only. The house 1 files carry more, and the pieces differ in purpose: some exist so we can understand the problem and will never be collectable in the target deployment; some map one-to-one onto what the planned submeters record. Every exhibit below carries a tag:

- **[insight only]** - for understanding the problem; not collectable in the target deployment and not usable for MVP experiments.
- **[collectable]** - the same quantity (or a close equivalent) can be recorded by the planned aggregate submeters.

Voltage rides along with every reading - [collectable]

The house 1 mains file stores three columns: `v0` active watts (used everywhere above), `v1` apparent power and `v2` mains voltage. The largest spike from the opening zoom, with voltage alongside:

voltage before event: 244.7 V | min during event: 241.0 V | sag 3.7 V (1.5%)



Reading it [collectable]. When the big load switches on, mains voltage dips visibly (the printed line quantifies the sag) and recovers when it switches off. Load events imprint on voltage even where power signatures are ambiguous - a free secondary signal on any meter that records it. It is also why two meters on the same circuit can disagree; the project standardises on active power.

Apparent vs active power - [collectable]

The mains file's second column, `v1`, is **apparent power** (VA). Its ratio to active power is a proxy for power factor — how much of the drawn current does real work. One representative day, sampled wherever active power exceeds 50 W:

apparent / active on 2013-10-03 (v0 > 50 W): p25 1.15 | p50 1.27 | p75 1.43 | above 1 in 100% of samples

Reading it [collectable]. The apparent/active ratio sits near 1.27 at the median (p25 1.15, p75 1.43) and is above 1 essentially always — the house carries reactive current and pulse loads that inflate VA without doing work. A meter that reports only apparent power overstates real consumption by that factor; the project standard (active watts) is the right call, and it is derivable on any meter that records voltage and current.

Button-press logs - human annotation - [insight only]

Alongside the power files sit 48 `channel\*\_button\_press.parquet` files: a resident pressed a button when switching an appliance on or off, logged as 0/1 toggles. No power at all - just timestamps and toggle state.

```
ch  label                toggle rows
--  -----
34  samsung_charger       2151
29  livingroom_lamp_tv    1647
26  livingroom_s_lamp2    1386
24  bedroom_ds_lamp       1318
45  childs_ds_lamp         1169
19  livingroom_s_lamp     1118
49  office_lamp2           968
33  utilityrm_lamp         924
43  data_logger_pc        777
48  office_lamp1          592
example - fridge (ch12): 335 toggles between 2013-03-20 11:41:24 and 2017-03-19 09:29:33
```

Reading it [insight only]. Even the fridge - used every day - earned only a few hundred toggles in 4.5 years; most channels got fewer. Manual labelling at this scale is a research luxury, not something a deployed pipeline can rely on: ground truth has to come from instrumentation or be inferred. The button logs show how little human-in-the-loop annotation actually yields.

When meters lie: implausible maxima - [insight only]

Small loads report sporadic spikes near 4 kW that the device cannot draw. The screen: channels whose median ON draw is under 150 W but which record readings above 2 kW in fewer than 0.1% of samples (the dishwasher's real 2 kW heater fails this filter in the right way — too many high readings to be noise):

```
ch  label                max W  p50 ON W  readings > 2 kW
--  -----
44  childs_table_lamp     3993   14.0     23
35  bedroom_d_lamp       3973   26.0      9
38  bedroom_chargers     3935    8.0      4
50  office_lamp3         3860    7.0      2
45  childs_ds_lamp       3795    8.0     19
31  kitchen_lamp2        3760   18.0      7
52  office_fan           3746   32.0      1
28  subwoofer_livingroom  3654   16.0     13
36  coffee_machine       3567   52.0     14
27  iPad_charger         3504    8.0     11
34  samsung_charger      3496    6.0      3
12  fridge               3323   89.0     13
48  office_lamp1         3220   14.0      5
15  hifi_office           3131    9.0      3
7   tv                  3109  103.0      2
32  kitchen_phone&stereo  2894   20.0      3
49  office_lamp2         2405   11.0      3
53  LED_printer          2242    6.0      2
23  kitchen_dt_lamp      2215   41.0      1
19  livingroom_s_lamp    2144   16.0      1
2   boiler              2060   12.0      2
21  gigE_&_USBhub        2058    9.0      1
37  kitchen_radio        2049  129.0      2
3   solar_thermal_pump   2048   48.0      2
47  battery_charger      2048   11.0      3
25 channels pass the spurious-spike screen
```

Reading it [insight only]. A table lamp tops the list at 3,993 W — the UK-DALE paper itself notes IAM readings up to ~4 kW are RF artefacts, and its authors filter values above 4 kW. 25 house 1 channels show the pattern: maxima far above any physical draw for the device, a handful of samples each. The fridge's 3,323 W spike (13 samples) is the same effect on a trusted channel. Lesson for the target deployment: screen `max / median-ON` before trusting any peak, and never let one spurious sample set a scaling range.

16 kHz V-I snippets - [insight only]

Beyond the power columns, the UK-DALE project published raw 16 kHz voltage-current capture windows for some houses. We hold none of it in `data/fnd/` (the V-I track under `research-logs/vi/` collects the equivalent idea for PLAID); it matters here as a pointer: high-rate V-I is what waveform-level appliance identification builds on — far beyond what an aggregate-only deployment can use, but the ceiling of what appliance identification could ever see.

The other four houses - same questions, different answers

House 1 got the deep dive because it is the richest. The fleet view: houses 2 and 5 also recorded a dedicated site meter (the quantity the target deployment records - **[collectable]**); houses 3 and 4 recorded their aggregate as an ordinary 6-second channel instead (**[insight only]**). Do the earlier answers transfer?

house	mains	span d	mean W	kWh	canonical present	fridge duty % (floor-aware)
house_1	yes	1501	343	12268	dishwasher, fridge, kettle, microwave, washing_machine	42.2
house_2	yes	176	299	1328	fridge, kettle, microwave, washing_machine	44.6
house_3	no	-	-	-	kettle	-
house_4	no	-	-	-	-	-
house_5	yes	137	572	1878	dishwasher, kettle, microwave	35.6
house_2 fridge (ch14 'fridge'): sensor floor 10 W -> plain-rule duty 100.0% (always-ON artifact); floor-aware duty 44.6%						

Reading it. The canonical targets are not uniformly present: houses 2 and 5 carry four of the five, house 3 only a kettle, house 4 a single combo channel that merges several appliances into one meter. Even the names differ across houses (`dish\_washer` vs `dishwasher`, `fridge\_freezer`, `washer\_dryer`) - label normalisation is part of the job. Where a fridge is separately metered, its duty differs from house 1's 42%: house 2's compressor runs ~45% of the time under the floor-aware rule — but the **plain rule reported 100%**, because that sensor idles at ~10 W and the rule's 5 W threshold never clears (printed under the table). Same appliance class, different home, different behaviour - and a threshold rule that must be checked against each sensor's idle floor. Multi-home datasets carry this friction, and models have to be robust to it.

Quirks & gotchas

- **Houses 3 and 4 have no dedicated 1 s mains file** — their aggregate is an ordinary channel (channel 1) at ~6-8 s cadence; useful as label-only context and low-rate practice, not for 1 s aggregate work.
- **Channel 1 is the aggregate itself** (label collision in `labels.dat`) — never sum it against the mains.
- Appliance channels run at **~6 s**: kettle inrush is smeared; episode timing carries ~6 s jitter. The mains is 1 s.
- **Duplicate labels** exist (two "light" channels) — treat them as separate meters of one appliance class.
- **Washer meter catches 14 s control draws** (18.7 episodes/day); long real cycles dominate its energy.
- **Mains holes: 10 d Jul 2016, 4 d Mar 2016** plus three evening dropouts — gate training rows around them.
- The mains also carries apparent power (v1) and voltage (v2); v0 (active W) is the NILM quantity, v2 allows voltage-sag studies.
- **Button-press logs** are sparse point events, not power series.
- `labels.dat` may list channels with no Parquet file (and vice versa); the inventory handles both.
- **ON-rule artifact on standing-draw sensors**: a meter idling at ~10 W (house 2's fridge) never clears the 5 W threshold, so the plain rule reads "always ON" (100% duty). The floor-aware rule (`thr = floor + max(5, 0.5\*(p50\_on - floor))`) fixes it; check every sensor's idle level before trusting duty.
- **Spurious ~4 kW readings**: 25 house 1 channels report sporadic RF artefacts up to 3,993 W (screen in Q8); the paper's authors filter above 4 kW.

- **Use-only meters:** iron (0.0% of the window), straighteners (0.2%), Hoover (0.7%), hair_dryer (1.0%) record only while in use — their energy covers recorded periods only, and OFF is indistinguishable from "logger parked".
- **No sentinel values:** UK-DALE writes plain 0 W for off (no negatives in any house 1 channel); ECO (notebook 04) uses -1 — check per dataset before applying `missing\_values`.
- **Times are UTC:** hour-of-day peaks are UTC; London runs UTC+1 in British Summer Time (late Mar-late Oct), so local-clock peaks shift one hour in summer exhibits.
- **Sub-2-minute dropouts are not OFF:** the paper treats IAM gaps under two minutes as non-recording; the same convention applies to the 6 s channels (a >2 min gap is missing data, not the appliance switching off).

Verdict — what is this dataset good for?

The primary training and evaluation substrate. 4.1 years of 1 s aggregate plus 53 labelled 6 s channels; all five canonical targets present with clean ground truth (34% of the bill between them, computed over the mains window); a stationary fridge anchor (Q4); mild repeatable seasonality (Q5); realistic overlap - 3.2% of minutes, but roughly half of events start inside a mix (Q6); a residual that is the meters' own draw, not hidden load (Q7); and step-size structure a model must survive (the kettle's median switch-on step is ≈ 225 W, not its 2.3 kW level, Q3). The gaps are documented and avoidable. Start every modelling question here.

A note on purpose: the appliance channels, button logs and voltage columns exist so we can understand what an aggregate-only deployment cannot see - they are **[insight only]**. The aggregate feed itself is **[collectable]**. What this dataset teaches about cadence, gaps, labelling and the un-metered residual is exactly what to get right when recording our own.

```
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    "mean_w": 342.5,
    "p50_w": 223.3,
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    "p99_w": 2710.7,
    "max_w": 8066.0,
    "energy_kwh": 12268.0,
    "steps_per_day_ge100w": 709.9
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  "gaps_house_1_mains": {
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```

```

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"fridge_weekly_duty": {
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  "p5": 35.0,
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  "weeks": 215
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"seasonality": {
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  "summer_mean_kwh": 208.1,
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"hourly_on_peak_hour": {
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  "washing_machine": 10,
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"simultaneity_two_plus_pct": 3.2,
"energy_coverage_pct": 86.5,
"energy_method": "timestamp-integrated over the mains window, per-sample carry-over capped at 3x cadence",
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  "p25_w": 56.0,
  "p75_w": 70.4,
  "night_mean_w": 58.0,
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  "r_labelled_mains": 0.975,
  "r_residual_mains": -0.06,
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  "nightly_floor_w": 111.0
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  "washing_machine": 51.0,
  "kettle": 52.0,
  "microwave": 56.1
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"dp_p50_w": {
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  "toaster": 1545,
  "microwave": 1323,

```

```

"kettle": 225,
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"defrost_share_of_on_pct": 2.9,
"voltage_sag": {
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"button_press_files_house_1": 48,
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    "-",
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```

```
    "house_5",
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},
"fridge_duty_pct_per_house": {
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  "house_3": "-",
  "house_4": "-",
  "house_5": "35.6"
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"lighting_longest_on_days": 7.84,
"boiler_longest_on_h": 770.7
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